Lam Fung Academy · LF Academy

Primary 5 · Semester 2 · 22nd Lesson · Student Handout

Fraction division (1)

Unit three · fraction÷integer ＋ fraction÷fraction · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Unit 3 + Modern Education 5L B Unit 11

Core traps:□ T3 Fraction division - forgetting to invert the divisor after the division sign (÷ becomes × reciprocal) · T9 The result is not reduced

SSPA association: 🔴 High frequency The sub-test must be taken every year, accounting for about 12-18% of Paper 1

Prerequisite knowledge: Course 7 (common denominators with different denominators) · Course 9 (multiplication of fractions, including mixed numbers × fractions)

Goals of this course:❶ Fraction ÷ whole number (=× reciprocal) ❷ Fraction ÷ fraction (reverse after division sign) ❸ Division of mixed numbers ❹ Division word problems

Student Name:	Class:	Date:	Time Spent:

I.Warm-Up Questions (total 4 question · 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Write the reciprocal of |||SEP|||. $\frac{2}{3}$ The countdown.	Basic	
2	Write what are the reciprocals of |||SEP|||? $\frac{1}{5}$ 、 $\frac{3}{4}$ 、 $\frac{5}{2}$ What are the reciprocals of ?	Basic	
3	Convert $2\frac{1}{3}$ into an improper fraction.	Basic	
4	Calculation: $\frac{3}{4} \times \frac{1}{2}$ = ? and reduce the results. (Review multiplication of fractions)	Basic	

II.Core Knowledge + Worked Examples

Knowledge Point 1: Fraction ÷ Integer🔴 SSPA

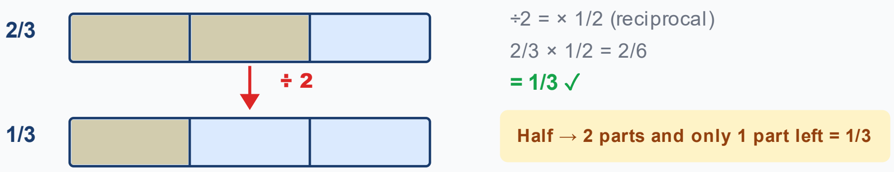
① **Core rule: fraction ÷ integer = fraction × (reciprocal of integer)**

② For example: $\frac{2}{3} \div 2 = \frac{2}{3} \times \frac{1}{2} = \frac{2}{6} = \frac{1}{3}$

③ Reciprocal of integer a = $\frac{1}{a}$ (remember: a = |||SEP|||, the reciprocal is $\frac{a}{1}$ ④ After division, it must be reduced! The numerator and denominator are divided by the greatest common factor (HCF) at the same time) $\frac{1}{a}$)

④ It must be reduced after division! Divide both the numerator and denominator by the highest common factor (HCF)

Fraction ÷ integer = fraction × reciprocal of integer



Example 1

calculate: $\frac{3}{4} \div 2 =$?

Example 2

calculate: $\frac{5}{6} \div 4 = ?$

Knowledge Point 1 Synchronization practice (write out: ① turn × reciprocal ② numeratormultiplynumerator·denominatormultiplydenominator ③simplify)

#	Question	Difficulty	Working Space
5	$\frac{2}{3} \div 2 = ?$	🌱	
6	$\frac{5}{8} \div 5 = ?$	🌱	
7	$\frac{4}{7} \div 3 = ?$	🌿	
8	$\frac{9}{10} \div 6 = ?$	🌿	

⚠ The most frequent error: ÷2 is treated as ×2 instead of × 12! Remember: ÷Integer = ×The reciprocal of that integer.

Knowledge Point 2: fraction÷fraction = × reciprocal 🔴 SSPA Must-Test

Core formula: $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}$

① The divisor after the division sign **must be inverted (the numerator and denominator are swapped)** ② The division sign becomes a multiplication sign ③ Then calculate by fraction multiplication ④ Must reduce Don't make a mistake: only the fraction after the division sign is inverted!

⑤ Make no mistake: only the fraction after the division sign is inverted!

The fraction after the division sign needs to be "reversed" - only the divisor is reversed

$2/3 \div \boxed{\frac{1}{4}}$ Turn over $\times \boxed{\frac{4}{1}} = 8/3 = 2⅔$

Rule: a/b ÷ c/d = a/b × d/c

Only the fraction after ÷ is reversed!

□ Trap detonation example 3 (the most important demonstration in this class)

calculate: $\frac{2}{3} \div \frac{1}{4} = ?$

❌ **Common mistakes (60% students)**

✅ **Correct solution**

$$\frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$$

Forgot to reverse the divisor of 14! Treat it directly as a multiplication sign.

$$\frac{2}{3} \times \frac{4}{1} = \frac{8}{3} = 2\frac{2}{3}$$

÷14 → ×41 (the fraction after the division sign must be reversed!)

🧠 **Tip: "Invert the division sign and turn it into a multiplication sign. The numerator is multiplied by the numerator and the denominator is multiplied by the denominator. The answer must be simplified!"**

Example 4

calculate: $\frac{3}{5} \div \frac{2}{3} = ?$

Knowledge Point 2 Synchronization practice (mustwrite out: ①÷→× ②divide number inversion ③simplify)

#	Question	Difficulty	Working Space
9	$\frac{1}{2} \div \frac{1}{3} = ?$	🌱	
10	$\frac{3}{4} \div \frac{1}{2} = ?$	🌱	
11	$\frac{5}{6} \div \frac{2}{3} = ?$	🌿	
12	$\frac{4}{5} \div \frac{3}{10} = ?$	🌿	

⚠️ High-frequency trap: Just remember "reverse" but also reverse the previous scores! For example, 23÷45 is incorrectly written as 32×54 — wrong! Only what follows ÷ is inverted.

Knowledge Point 3: Division with Mixed Fractions 🔴 SSPA Advanced

Standard three-step method:

① **mixed fraction → improper fraction**(integer × denominator + numerator)

② **Reverse the divisor after the division sign**, the division sign becomes the multiplication sign

③ **Multiply the numerator by the numerator and the denominator by the denominator**→ Reduction → Convert the improper fraction to a mixed fraction ision of mixed numbers: first convert to improper fraction → then reverse → then multi

)Convert to improper fraction $1\frac{1}{2} \to 3/2$ → ②Reverse divisor $\div 2/3 \to \times 3/2$ → ③Multiplication calculation $3/2 \times 3/2 = 9/4 = 2\frac{1}{4}$

①	②	③	④
Convert Improper Fractions Convert all mixed numbers into improper fractions	**divisor reversal** ÷The following fraction is reversed	**Multiplication calculation** Numerator × Numerator Denominator × Denominator	**Reduction·Return** Reduction + Improper Fraction → Mixed Fraction

Example 5

Calculate: $1\frac{1}{2} \div \frac{2}{3} = ?$

Example 6

Calculate: $2\frac{1}{3} \div 1\frac{1}{2} = ?$

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
13	$1\frac{1}{4} \div \frac{1}{2} = ?$	🌿	
14	$2\frac{1}{2} \div 1\frac{1}{3} = ?$	🌳	
15	$1\frac{2}{3} \div \frac{3}{4} = ?$	🌳	

❌ **Trap: Mixed fractions are reversed before they are false.**

$$1\frac{1}{2} \div \frac{2}{3} = \frac{1}{2} \times \frac{3}{2}$$ ❌

Just separate the numerator and denominator of 1½ – big mistake!

✅ **Correct: Convert improper fractions first**

$$\frac{3}{2} \times \frac{3}{2} = \frac{9}{4} = 2\frac{1}{4}$$

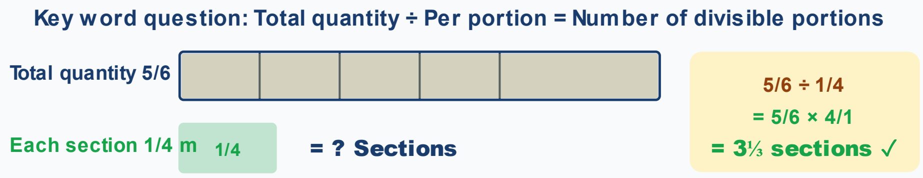
$1\frac{1}{2} = 3/2 \to 3/2 \div 2/3 = 3/2 \times 3/2 = 9/4$

Knowledge Point 4: Fraction Division Application Questions 🔴 SSPA Required Exam

Four-step method to solve the problem: ① **Circle keywords** ("Share equally" "Each..." "How many servings can..." = division)

② **Column formula** (Total amount ÷ per serving = number of servings; Total amount ÷ number of servings = per serving)

③ **Calculate according to the division steps** ④ **Write a complete answer** (Step points will be deducted if you do not write an answer!)



Example 7

$\frac{3}{4}$ A cake is divided equally among 5 people. What fraction of the cake does each person get?

Example 8

A ribbon is $\frac{5}{6}$ meters long, cut into sections every $\frac{1}{4}$ meters. How many pieces can it be cut into?

Knowledge Point 4: Synchronized practice (column → calculate → answer sentence)

#	Question	Difficulty	Working Space
16	$\frac{1}{2}$ A pancake is divided equally among 3 people. What fraction of the pancake does each person get?		
17	A bottle of juice contains $\frac{4}{5}$ liters. Pour one cup for every $\frac{2}{5}$ liter. How many cups can be filled?		
18	A rope is $2\frac{1}{2}$ meters long and cut into sections every $\frac{1}{2}$ meter. How many pieces can it be cut into?		

III. Lesson Layered Synchronization Practice

Basic layer (total 4 questions, everyone must do)

#	Question	Difficulty	Working Space
19	$\frac{3}{5} \div 3 = ?$	🌱	
20	$\frac{4}{7} \div 2 = ?$	🌱	
21	$\frac{3}{8} \div \frac{1}{4} = ?$	🌱	
22	$\frac{5}{6} \div \frac{5}{12} = ?$	🌿	

Advanced layer (total 3 questions, 🚶🚀 choose do)

#	Question	Difficulty	Working Space
23	$\frac{7}{9} \div \frac{2}{3} = ?$	🌿	
24	$1\frac{1}{5} \div \frac{3}{5} = ?$	🌿	
25	$\frac{9}{10} \div \frac{3}{5} = ?$	🌿	

🌳 challenge layer (total 2 questions, 🚀 choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
26	$\frac{3}{4} \div \frac{2}{3} \div \frac{1}{2}$ =? (Continuous division of three fractions - calculate the first two first, then ÷ the third result)	🌳	
27	$2\frac{1}{4} \div 1\frac{1}{2} = ?$	🌳	

IV.⛰️ Ultimate challenge area

#	Question	Difficulty	Working Space
⛰️ 1	A rope is $3\frac{1}{2}$ meters long. Use half of it first, and then cut the remaining $\frac{1}{4}$ meters into a section. How many pieces can it be cut into? (You need to find the remaining length first, and then ÷ the length of each paragraph)	⛰️	
⛰️ 2	A box of candy weighs $\frac{3}{4}$ kilograms, and each $\frac{1}{8}$ kilogram contains a small package. After filling 3 bags, how many bags of candy can be left?	⛰️	

advancedapplicationquestion (🧗🚀 choose do, SSPAPaper 2commonquestion type)

#	Question	Difficulty	Working Space
A1	$\frac{2}{3}$ Kilograms of sugar, one pack per $\frac{1}{9}$ kg. How many bags can be packed?	🌿	
A2	The original water bottle contains $\frac{3}{4}$ liters. After drinking $\frac{1}{3}$ liters, pour the remaining water into 2 cups. How many liters of water are in each cup?	⛰️	

V. Class afterhomework

Basic must doquestion (total 6 questions, mustwrite out complete step: ÷→×→reverse→simplify)

#	Question	Difficulty	Working Space
H1	$\frac{2}{5} \div 2 = ?$	🌱	
H2	$\frac{3}{8} \div 3 = ?$	🌱	
H3	$\frac{1}{2} \div \frac{1}{4} = ?$	🌱	
H4	$\frac{3}{4} \div \frac{1}{3} = ?$	🌿	
H5	$\frac{7}{10} \div \frac{2}{5} = ?$	🌿	
H6	$1\frac{1}{2} \div \frac{1}{2} = ?$	🌿	

For advanced, choose doquestion (total 2 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H7	$2\frac{1}{3} \div 1\frac{1}{4} = ?$	🌳	
H8	There are $\frac{3}{5}$ liters in a bottle of orange juice. Pour a small glass for every $\frac{3}{20}$ liter. After filling 2 cups, how many more cups can be filled?	⛰	

VI. The Lessoncorecommon errorsummary

✅ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🌱basic questions independently ☐ I can challenge 🌿advanced questions

☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall ☐ Solve 🌱basic questions independently (100% correct)

☐ Challenge 🌿Advanced questions (80%+ correct) ☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	**÷integer is treated as×integer** : $\frac{3}{4} \div 2 = \frac{6}{4}$	$\div 2 = \times\frac{1}{2}$, not ×2! The reciprocal of an integer is 1/integer
2	**÷Fraction forgetting to reverse the divisor** : $\frac{2}{3} \div \frac{1}{4} = \frac{2}{12}$	$\div\frac{1}{4} = \times\frac{4}{1}$, only the one after the division sign is inverted!
3	**Reverse the previous fraction as well** : $\frac{2}{3} \div \frac{4}{5} = \frac{3}{2} \times \frac{5}{4}$	Only fractions after ÷ are inverted! The front is unchanged, the back is reversed.
4	**The answer is not reduced** : $\frac{6}{12}$ as the final answer	Must be reduced to the simplest fraction (numerator and denominator ÷ HCF)
5	**Improper fractions are not converted to mixed fractions**	Numerator > Denominator → Convert into mixed numbers (subject to the requirements of the division test)
6	**Mixed fractions are divided directly by** ÷without conversion: 1½÷2	You must first convert to an improper fraction (1½ = 3/2), and then follow the division steps
7	**Application question "number of copies vs. each copy" confusion**	Total amount ÷ amount per serving = number of servings; total amount ÷ number of servings = amount per serving. Distinguish clearly!